

Derivation of Dark Matter, Dark Energy and MOND, as Features of Space-Time

-- Complete Solution of the Dark Matter and Dark Energy Problems Through Einstein's Field Equation --

P. H. Handel^{1*} and K. E. Splett^{2†}

Affiliations: ¹Department of Physics and Astronomy, University of Missouri-St. Louis & Center for Nanoscience; St. Louis MO 63121 USA. ²University of Missouri-St. Louis Visiting Scholar 2017-2022; [†]Present address: Petersroda 06809 Germany. *Corresponding author phasenoisehandel@gmail.com, handel@umsl.edu, erikasplett@yahoo.de;

Abstract. We solve the problem of Dark Matter (DM) and Dark Energy (DE), from Einstein's field equation (EE), determining their density at any point of the observable universe (OU), agreeing with all known observations. From EE we calculated the metric and DM, about five times the ordinary matter, earlier [Handel and Splett, Foundations of Physics, **53** (5) 2023, DOI 10.1007/s10701-023-00705-x] as the energy of the coherent gravitational field of the whole OU, distributed on lower-level structures proportional to the classical gravitational field g . This g is calculated without DM or DE. Physically solving EE, we now derive MOND, the Modified Newtonian Dynamics, and derive simple universal analytical formulas exactly agreeing with observations, for DM density at all OU levels, for DE density, and for the cosmological constant Λ . This yields our "DM key" for DM concentrations, $\rho_D = Qg$. The constant $Q = 1.39 \cdot 10^{-16} \text{ gs}^2/\text{cm}^4$ is the DM concentration accreted from the OU, on the g of lower-scale structures, per unit local Newtonian g . We exactly derive analytically

from EE: the galactic rotation curves, the observed extended galactic halos, the Tully-Fisher relations and MOND (generalized). Our key easily calculates analytically the distribution of DM in galaxy clusters, superclusters, colliding clusters, planetary systems, wide binary stars, or on earth as **134 grams/km³**. We derive Λ as $1.08 \cdot 10^{-56} \text{ cm}^{-2}$ and $\rho_L = 5.8 \cdot 10^{-30} \text{ g/cm}^3$, with error $< 3\%$, verified by self-consistence. This theory replaces earlier hypothetical DM models.

Keywords: Astrophysics, Cosmology, Gravitation, Dark matter, Dark Energy, Cosmological Constant, General relativity, Interdisciplinary physics, Fundamental 1/f noise.

1. Dark Matter as the Gravitational Field Energy of the Observable Universe

According to the well-known Λ CDM (Lambda Cold Dark Matter) model of Astrophysics, about 26% of the mass of the universe is DM,^{1,2} usually assumed to be some hidden particles¹, interacting only gravitationally. Ordinary matter (OM), including gas, radiation, all particles,

plasma, dust, stars, planetary systems, galaxies and clusters, represents about 5%, mainly baryonic matter. The rest of 69% is considered^{1,2} to be DE and is described in Einstein's field equation (EE) by the cosmological constant, observed to be $\Lambda=1.088 \cdot 10^{-56} \text{ cm}^{-2}$, or by a DE density corresponding to a matter density $\rho_\Lambda = \Lambda c^2 / 8\pi G = 5.96 \cdot 10^{-30} \text{ g/cm}^3$. The DE density ρ_Λ , is also generally considered to be of unknown nature. It is equivalent to a negative pressure that generates a "gravitational repulsion." This is needed to explain with EE the acceleration of the Hubble expansion of the universe, first observed in 1999. The DE is considered responsible for reducing the curvature of the *perceived* OU to zero, causing it to be quite flat, with a curvature parameter $\Omega_K = 0.0007(19)$. This was obtained in a 6-Parameter 2018 Λ CDM cosmology fit², to the Planck telescope data on the cosmic microwave background temperature, polarization, and lensing. The bracket (19) shows the uncertainty in the last digits of Ω_K . Since the central value of this most recent result is positive, we consider the perceived universe likely to be open and practically flat on the largest scale. We remind that $\Omega_K < 0$ corresponds to a finite, spherical geometry of the universe, $\Omega_K > 0$ to a hyperbolic open geometry, and $\Omega_K = 0$ to a flat (Euclidian) open geometry. The last two cases correspond to an open, mostly unobservable, universe.

In a previous paper³ we explained DM in principle as energy of the coherent gravitational field of the universe (Eventually divided by c^2 to get it in grams). *We defined DM as the additional matter needed at all length scales to explain, using classical Newtonian physics, the observed gravitational field.*³ *Rather than trying to replace the notion of DM outright, with its expression in terms of the metric tensor, or declaring that it does not exist, we found it reasonable to keep using this re-defined notion of DM.* The gravitational energy of the universe, i.e. universal DM, was calculated in four ways³, both in low-order GR (quadratic in the deviation $h_{\mu\nu}$ from the Minkowski-Euclidian metric), and classically. It was found each time to be of the same order of magnitude as the OM energy. With our new definition, we showed that *DM would largely become its own source³ in EE, and could indeed self-generate more DM* to be about five times larger than the OM, at the level of the OU. This was shown by using the GR quasi-tensor $t_{\mu\nu}$ of the gravitational field defined by Weinberg⁴ pp. 166-167, Eqs. 7.6.14-7.6.15. At the local, lowest scales of the OU, as we will show in the present paper, the simple NA is applicable, with increasing GR corrections at progressively larger scales. These corrections are calculated automatically by our "DM key" algorithm derived in Sec. 2 below. Our analytical results for galactic rotation curves and DM halos agree perfectly with all astrophysical observations. The Tully-Fisher

relations are derived in Sec 3, the DE in Sec. 4. The original derivation of DM, based on a simple electromagnetic analogy, is shown in Secs. 5 and 6 to yield about 5 times the OM. Sec. 7 derives MOND for the first time with our DM key, essentially from the original EE, without hypothetical DM particles. In Sec. 8 we consider the DM effects in wide binary stars, separated by distances of light years. We discuss the meaning of our physical solution of EE and of the DM, or DE and Λ in Sec 9.

2. The Rotation Speed of Stars in Galaxies

Let us imagine a rigorous solution of the non-linear EE for all coupled levels of the universe, that would define the metric of space-time everywhere, implying the gravitational field and its energy, where the energy mainly represents the DM. The question arises then: *how does the observed DM, about 5 times larger than OM at the level of the OU, manifest itself (in this imagined solution) as gravitational energy or DM in lower-level structures of the universe, i.e. in planetary systems, galaxies, and clusters thereof?*

We would expect the gravitational field of the universe, denoted in simplistic way with the symbol g' , to add energy (or DM) with a density dependent on both g and g' , to the much smaller gravitational field g of lower-level structures. This smaller g is generated in the local system of reference (just as $g=9.8 \text{ m/s}^2$ on earth). Indeed, EE implies a self-interaction of the gravitational field of the OU everywhere, at all levels.

This happens because the gravitational field has energy and therefore mass, but at the same time it is also the force acting on each mass unit. Thus, it also acts on its own mass. In the solution of the nonlinear EE for the OU, we therefore expect to find a "universal" energy density contribution ρ_D , which is a function $\rho_D = f(g, g')$ of g and g' , in addition to the local Newtonian energy density ρ of the smaller local field $g \ll g'$ that is generated in lower-scale structures.

By expanding $f(g, g')$ in series around $g=0$, with $g \ll g'$, we retain the first-order term

$$\rho_D = \rho_D(0) + Qg = Qg, \quad (1)$$

where Q is a function of g' , with $\rho_D(0)=0$. We are interested in the term Qg , which describes the accretion of universal gravitational energy in mass units, per unit volume on the classical local gravitational field g , measured in the rest frame of the lower-level structure considered, e.g., a cluster of galaxies, a galaxy, a star, or a planet. It does not matter where the galaxy is in its cluster, where the star is in the galaxy, or where the planet is in the planetary system, as long as they are in free motion, in "free fall."

The result in Eq. (1) can also be understood on a simple physical basis. Considering g' as a measure of the gravitational field of the universe, g'^2 can symbolize a measure of the energy or mass density of the gravitational field of the universe, i.e. density of DM. *The field g of a lower-scale structure is the force acting on any mass unit* and will therefore attract

universal DM with a force density proportional to gg'^2 . We may thus *again expect the density of DM* attracted into a place in a lower-scale structure (where the locally generated field is g) to be Qg , i.e., proportional to the attraction force, or to g , with $\rho_D(0)=0$. So we consider Q to be proportional to g'^2 . *Introducing the proportionality factor λ , we obtain*

$$\rho_D = \lambda gg'^2 = Qg. \quad (2).$$

The important Ansatz $\rho_D = Qg = \lambda gg'^2$ as the DM key for distribution and accretion of universal DM into lower-scale structures was thus obtained in two ways, Eqs. (1) and (2). The second way is physical and less rigorous, but shows that $\rho_D(0)=0$. We use both Q and $\lambda g'^2$ as equivalent, interchangeable symbols. We shall see below that this simple Ansatz $\rho_D = Qg$, *our universal DM key*, is applicable with excellent results in any lower-scale structures.

We now apply our DM key to find the accretion of universal DM onto the gravitational field generated by a galaxy in its system of reference. The rotation speed v of a star of mass m , predicted by Newtonian physics in a galaxy with a total baryonic mass M , concentrated uniformly at $r < a$ in the center, as we assume here, is given for $r > a$ by

$$mv^2/r = GmM/r^2; \quad v = (GM/r)^{1/2}. \quad (3)$$

In Fig. 1 below, v should follow the dotted lower curve A, but is observed to follow the upper red curve B, showing an almost constant value of v for $r > a$.

We expect DM generated at the level of the universe to be attracted by the locally generated classical gravitational field $g = GM/r^2$. The DM is attracted proportional to gg'^2 and added to the existing baryonic matter concentration according to our key $\rho_D = Qg = GMQ/r^2$ (for $r \geq a$), the key introduced in Eq. (1). For $r \leq a$ we obtain $\rho_D = Qg = GM(r^3/a^3)Q/r^2 = GMQr/a^3$. The total mass $M_<$, found up to the distance $r \geq a$ from the galactic center, is then

Fig. 1: After reaching a maximum close to the Newtonian prediction A, the rotational speed of stars follows the red curve B and remains roughly constant with distance r . By Phil Hibbs - his work in Inkscape 0.42, Public Domain, <https://commons.wikimedia.org/w/index.php?curid=365013>.

$$\begin{aligned} M_< &= M + GQM \left[\int_0^a r^3/a^3 + \int_a^r \right] (4\pi/r^2) r^2 dr \\ &= M + GQM \int_0^a (4\pi r^3/a^3) dr + 4\pi GQM(r-a) \\ &= M - 3\pi QGMa + 4\pi QGMr. \end{aligned} \quad (4)$$

Thus, for $r \geq a$, the concentration of DM decreases like $1/r^2$, *but the cumulative DM increases proportional to r , creating an extended DM halo of the galaxy. The final gravitational field g_f resulting from the Gauss theorem from Eq. 4 at $r > a$ is thus*

$$g_r = M_{<}/r^2 = MG[1-3\pi QGa]/r^2 + 4\pi QG^2M/r. \quad (4')$$

For $r \leq a$ we obtain, with a DM concentration $\rho = QGMr/a^3$,

$$M_{<} = Mr^3/a^3 + 4\pi QGM \int_0^r r^3/a^3 dr \\ = Mr^3/a^3 + \pi QGM r^4/a^3, \quad (5)$$

showing the complete absence of any DM concentration cusp in the galactic center. A cusp had been predicted by Λ CDM simulations with hypothetical-particle-DM.

The rotation speed of stars in galaxies is given then, for $r \geq a$, by

$$v^2 = G(M_{<})/r = GM[(1+\pi QGa)/r + 4\pi QG(1-a/r)] \\ = GM[(1-3\pi QGa)/r + 4\pi QG]. \quad (6)$$

For the first time, this now reveals a constant v^2 of $4\pi QMG^2$ even for arbitrarily large distances r , as indicated by curve B in Fig.1. This will be shown again in more detail in Eq. (16) of Sec. 3. We expect this constant value of v to continue far beyond the distance up to which the Doppler effect in the 21 cm spectral line of the rotating Hydrogen ring of the galaxy allows us to measure it (a few 10^5 ly). The rotation with constant v should not stop before the gravitational field changes sign, pointing to a neighboring galaxy (about $1.25 \cdot 10^6$ ly for the Milky Way).

This startling prediction was recently verified on both early and late types of *isolated* galaxies, from their observed weak gravitational lensing, by Mistel et. al⁵. Their results showed no decrease in circular velocity up to distances of 1 Mpc = 3,261,564 ly, well beyond the predictions of all existing DM models.

These new measurements clearly verify and validate our GR based theory, eliminating all hypothetical DM models that have been proposed.

For $r \leq a$, dropping the second integral in rectangular brackets, we obtain from Eq. (4)

$$v^2 = G(M_{<})/r = GM[r^2/a^3 + G\pi Qr^3/a^3], \quad (7)$$

consistent with the absence of a central DM cusp.

Eqs. (6) and (7) analytically reproduce the features of curve B in Fig. 1, with continuity at $r=a$. Indeed, they yield a continuous velocity curve, with a likely discontinuity of the derivative at $r=a$, depending on the value of a .

In Sec. 3 we apply our DM key to the derivation of the Tully-Fisher relation. Then we use the experimentally well-known constant present in this relation to determine the value of the proportionality factor $Q = \lambda g'^2$. The value of $Q = \lambda g'^2 = 1.39 \cdot 10^{-16} \text{ gs}^2/\text{cm}^4$, obtained in Sec. 3, yields $3\pi QGa = 1.083$. This is slightly larger than 1 and for $r > a$ yields a small, positive derivative of v^2 , rapidly decreasing to zero, as seen in curve B, Fig. 1.

Using our DM key we have also calculated³ the DM in the solar system and found that it yields a negligible correction on the order of 10^{-9} to the mass of the sun at the orbit of the earth. Calculation also showed a solar mass correction of 10^{-4} at a distance of 2 light years (ly) from the sun, in the small DM halo of the sun, almost half the distance to the nearest star Proxima

Centauri. We also considered qualitatively the DM effects in the Bullet cluster, clusters, and super-clusters of galaxies in general, in startling qualitative agreement with observations. Finally, we elucidated how DM is dominant at higher scales of the universe and becomes its own source, appearing on both sides of EE. This self-interaction of the largest scale gravitational field, of DM leads to its own clumping tendencies and has led, early in the evolution of the universe, to the formation of the cosmic web. The latter is a 3-dimensional system of DM branches loosely filling the universe with a web-like pattern, discovered in the last few decades. It is decorated with galaxies, their clusters and superclusters, with very large voids between branches. The idea of self-interaction of the gravitational field, seen in interactions between gravitons, was also connected⁶ earlier with the problem of DM.

3. Derivation of the Tully-Fisher Relations

These relations⁷ state that *the baryonic mass M of a galaxy is proportional to about the fourth power of the peripheral speed v of the OM in it:*

$$M/M_{\odot} = 58(v/1\text{km}\cdot\text{s}^{-1})^{3.97}, \quad (8)$$

were $M_{\odot}$ is the mass of the sun. In Eq. (4') the first term in rectangular brackets is proportional to $1/r$ and decreases to zero for $r \ll D$. We can neglect it at larger r values and in general. Thus, by substituting for r in Eq. (4') the total radial extension $L_g = D$ of the galaxy including the halo, using the Gauss Theorem $\int \mathbf{g} \cdot d\mathbf{S} = 4\pi G M_t$, we obtain a

good approximation of the total (OM + DM) mass M_t of the galaxy

$$M_t \approx 4\pi G Q M D, \quad (9)$$

in agreement with Eq. (4). Its total gravitational energy in the NA is thus

$$G(M_t)^2/2D = \gamma M c^2. \quad (10)$$

This equation compares this gravitational energy with the rest energy Mc^2 of OM in the galaxy and introduces γ as their ratio. The maximal rotational speed, mainly of hydrogen, at the periphery of the galaxy is then given by

$$v^2 = G M_t / D = [2\gamma G M c^2 / D]^{1/2}. \quad (11)$$

On the right-hand side we have substituted M_t from Eq. (10). From Eq. (11) *we derive the Tully-Fisher relations in this form:*

$$v^4 = 2\gamma G M c^2 / D \quad (12)$$

in which the exponent 3.97 is replaced by 4. ***This is the first general derivation of the Tully-Fisher relations from first GR (and local NA) principles, to our knowledge.***

Eq. (12) allows for a determination of the coefficient γ by comparing Eqs. (8) and (12), in CGS units, with $D = 5 \cdot 10^5$ ly $= 4.7 \cdot 10^{23}$ cm,

$$\begin{aligned} 2\gamma &= D(1\text{km/s})^4 / 58 G c^2 M_{\odot} \\ &= 4.7 \cdot 10^{23} \cdot 10^{20} / 58 \cdot 6.7 \cdot 10^{-8} \cdot 9 \cdot 10^{20} \\ &\cdot 2 \cdot 10^{33} = 6.8 \cdot 10^{-6}. \end{aligned} \quad (13)$$

This yields $\gamma = 3.4 \cdot 10^{-6}$ for a typical galaxy used in the Tully-Fisher studies.

Now Eq. (13) allows us to derive the quantity $Q = \lambda g'^2$ in Eq. (1), which describes the accretion of DM generated at the level of the observable universe into lower-scale structures. Indeed, by eliminating M_t from Eqs. (9) and (10), we obtain from Eq. (13),

taking the baryonic mass of an average galaxy to be $M=M_g=1.4 \cdot 10^{43}g$,

$$\begin{aligned} Q &= (c/4\pi)(2\gamma/MDG^3)^{1/2} = (3 \cdot 10^{10}/4\pi) \\ &[6.8 \cdot 10^{-6}/1.4 \cdot 10^{43} \cdot 4.7 \cdot 10^{23} \cdot (6.7)^3 \cdot 10^{-24}]^{1/2} \text{ gs}^2/\text{cm}^4 \\ &= (2.39 \cdot 10^9)[3.4 \cdot 10^{-51}]^{1/2} \\ &= 1.39 \cdot 10^{-16} \text{ gs}^2/\text{cm}^4. \quad (14) \end{aligned}$$

This defines the coefficient of the universal DM accretion key, determining the amount of DM found in galaxies, their clusters, and in all lower-scale structures, such as planetary systems, binary stars, or planets. As shown in Sec. 4, this also allows for the first calculation of DE and of the cosmological constant. From Eqs. (13) and (14) we see that $Q \equiv \lambda g'^2$ does not depend on the value of D.

Furthermore, Eq. (14) fixes the relative value of the DM terms with Q in Eqs. (4)-(7) as was claimed above, directly below Eq. (7). At $r=a$, the DM terms in rectangular brackets in Eq. (6) together yield one third of the DM term in round brackets. These DM terms ($3\pi GQ$ and $4\pi QG$) are to be compared with $1/a$, which comes from the first term in round brackets, the baryonic contribution. Approximating a in Eq. (6) with the size of the main central baryonic mass of a typical galaxy in the mentioned Tully-Fisher study, of $a=1.3 \cdot 10^4 \text{ ly}$ (about the size of the central bar of a spiral galaxy), using Eq. (14) we obtain

$$\begin{aligned} 3\pi QGa &= 9.42 \cdot 1.39 \cdot 10^{-16} \cdot 6.7 \cdot 10^{-8} \cdot 1.3 \cdot 9.5 \cdot 10^{21} \\ &= 1.083. \quad (15) \end{aligned}$$

This is slightly larger than 1, yielding a slightly negative first term for v^2 in Eqs. (6)

and (16), quickly decreasing like $1/r$ in absolute value. Therefore, for $r \geq a$, from Eq. (6) we obtain:

$$\begin{aligned} v^2 &= GM[(1-1.083)/r + 1.444/a] \\ &= GM[(1-1.083)/r + 1.17 \cdot 10^{-22} \text{ cm}^{-1}] \\ &= M\{-5.56 \cdot 10^{-9}/r + 7.84 \cdot 10^{-30} \text{ cm}^{-1}\} \text{ cm}^3/\text{gs}^2. \quad (16) \end{aligned}$$

This yields a small, fast decreasing, positive derivative $(d/dr)(v^2)$ at $r > a$, in very good agreement with the observed curve B in Fig. (1). The slight undulation and, later, small bulging on curve B can be related to the presence of stars at distances larger than a . More important is the presence of the second term in the rectangular brackets ($1.17 \cdot 10^{-22} \text{ cm}^{-1}$) and curly brackets ($7.84 \cdot 10^{-30} \text{ cm}^{-1}$) of Eq. (16). This term does not decrease with r , in agreement with the practically constant extension of observed curve B into arbitrarily large values of r .

For $r < a$ we obtain from Eq. (7)

$$v^2 = GM[r^2/a^3 + 2.92 \cdot 10^{-23} \text{ cm}^{-1} r^3/a^3]. \quad (17)$$

This result ignores at $r \ll a$ the presence of a black hole at $r=0$ in most galaxies.

The consistency of the exact results obtained with our DM key verifies our physical approximation of unknown GR results, expected from EE. Moreover, as our derivations show, our theory does not contain any free, adjustable, arbitrary parameters or "fudge factors."

The exact calculation of DM density at any OU point, including derivation of DM content in any volume on the earth (134 g/km^3), could also allow direct interferometric verification with recently developed, ultra-sensitive devices.⁸

4. Derivation of the Dark Energy

We start from our perception that the OU appears to be flat at the largest scale, with a known² curvature parameter $\Omega_K = 0.0007(19)$, very close to zero, as mentioned above. We thus consider a "bumpy observable universe" flat on the larger scale. In this universe, the "dark energy," described by the cosmological constant Λ in EE, yields a mass density of $\rho_\Lambda = 5.96 \cdot 10^{-30} \text{ g/cm}^3$. This Λ corresponds to the energy density estimated from observations for a flat universe. It implies compensation for the average curving effect of matter and field, i.e. of OM and DM. *We derive this dark energy density and its corresponding mass density ρ_Λ here with the help of our DM key, considering an equivalent, homogeneous OU, neglecting small-scale inhomogeneities.*

For an analogy, let's compare the universe to a 2-D square lattice of identical, well-separated, galactic disks of radius L_g , on an infinite plane. Consider the average compensation of the curvature effects in each of the units or "tiles," representing a galaxy, including the appropriate average empty space surrounding it.

In three dimensions, this would correspond to a "face-centered cubic" (FCC, close-packed) lattice of galaxies, each of baryonic mass M_g , and radius L_g , with a large surrounding associated empty space of size $L \gg L_g$. This yields a total volume of $4\pi L^3/3$ per galaxy. This surrounding associated space accounts for the many large voids present between the branches of the

cosmic web. The actual galaxy size L_g is much smaller than the lattice constant $1.81 \cdot L$ of the primitive (rhombohedral, parallelepipedal) cell of this FCC lattice. The FCC conventional *cubic* unit cell has an edge of $2.56L$ and contains 4 primitive cells, each with one lattice point.

We can display this compensation of opposing effects on the geometric curvature of the universe in terms of the masses (or equivalent energies) that cause them. *On one hand there is* the total average galactic mass, baryonic (OM) and DM, *on the other is* the dark energy $\rho_\Lambda \cdot L^3$. Here $L^3 > L_g^3$ is the total average volume, including the associated portion of voids, corresponding to a galaxy.

The first two terms in the last form of Eq. (4) almost cancel each other (like 1-1.083) and quickly decrease in importance as r increases, if compared to the last term. To describe this curvature balance at the level of individual galaxies, we retain only the last term in Eq. (4) with $r = L_g$, which is the average radius of a galaxy, including its effective DM halo. The curvature, caused by this term in EE, is compensated by Λ , i.e. by the curvature effect of the dark-energy-equivalent mass concentration ρ_Λ multiplied by the volume $4\pi L^3/3$. We will display this local compensation as a compensation of the corresponding masses:

$$4\pi Q M_g L_g - \rho_\Lambda 4\pi L^3/3 = 0. \quad (18)$$

Eq. (18) yields

$$\rho_\Lambda = 3Q M_g L_g / L^3. \quad (19)$$

With the help of our universal DM key, its coefficient $Q = 1.39 \cdot 10^{-16} \text{ gs}^2/\text{cm}^4$, and with

$G=6.67 \cdot 10^{-8} \text{ cm}^3/\text{gs}^2$, we obtain

$$\rho_{\Lambda} = 3 \cdot 6.67 \cdot 10^{-8} \cdot 1.39 \cdot 10^{-16} \text{ cm}^{-1} M_g L_g / L^3 \\ = 2.78 \cdot 10^{-23} \text{ cm}^{-1} \cdot M_g L_g / L^3. \quad (20)$$

We introduce the average baryonic mass of a galaxy in the OU, also used in Eq. (14),

$$M_g = 1.4 \cdot 10^{43} \text{ g}. \quad (21)$$

As previously mentioned, the value of the outer radius L takes into account the many large voids present between the branches of the cosmic web. This web may contain about $2 \cdot 10^{12}$ galaxies. L is obtained by scaling down the radius R_u of the OU by a factor of $(2 \cdot 10^{12})^{1/3}$, yielding

$$L = (2 \cdot 10^{12})^{-1/3} \cdot 4 \cdot 10^{28} \text{ cm} = 3.175 \cdot 10^{24} \text{ cm} \\ = 3.34 \cdot 10^6 \text{ ly}; \quad L^3 = 3.2 \cdot 10^{73} \text{ cm}^3 \quad (22)$$

Considering an average effective radius $L_g = D$ of the galactic disk, including the main part of the rotating hydrogen belt and DM halo, $L_g = 5 \cdot 10^5 \text{ ly} = 4.75 \cdot 10^{23} \text{ cm}$, we obtain from Eq. (20)

$$\rho_{\Lambda} = 2.78 \cdot 10^{-23} \cdot 1.4 \cdot 10^{43} \cdot 4.75 \cdot 10^{23} / 3.2 \cdot 10^{73} \text{ g/cm}^3 \\ \text{cm}^3 = \underline{\underline{5.8 \cdot 10^{-30} \text{ g/cm}^3}}. \quad (23)$$

This is in very good agreement with the observed dark energy density value known to be $\rho_{\Lambda} = 5.96 \cdot 10^{-30} \text{ g/cm}^3$. The error is less than 3%. The corresponding cosmological constant is thus

$$\Lambda = (8\pi G/c^2) \rho_{\Lambda} = 8 \cdot 3.14 \cdot 6.67 \cdot 10^{-8} \\ \cdot 5.8 \cdot 10^{-30} \text{ cm}^{-2} / 9 \cdot 10^{20} = \underline{\underline{1.08 \cdot 10^{-56} \text{ cm}^{-2}}}. \quad (24)$$

This, again, is very close to the known value of $\Lambda = 1.088 \cdot 10^{-56} \text{ cm}^{-2}$. This is a first simple derivation of the cosmological constant, based, in fact, on Einstein's equation, and on its local Newtonian approximation, leading to our "DM key." This is a first solution of the "cosmological constant problem."

These very close results are very remarkable, but also fortuitous in their extreme precision (error $< 3\%$), given the uncertainties in L_g , L and M_g . It is important that the order of magnitude our results is correct. It shows how well our "DM key" works to approximate and anticipate the results from the nonlinear EE. Indeed, EE is known to be prohibitively difficult to solve exactly, even more so for all interacting scales of the universe.

Our DM key also describes the large DM effects observed in collisions between clusters of galaxies, as seen in the well-known Bullet cluster and in the recently studied MACS J0018.5+1626 clusters⁹ merger. In both cases, the DM seems to outrun the intra-cluster hot ionized gas that provides most of the baryonic mass. Our DM key yields a DM distribution proportional to the classical g , centered on the hot gas, but which is zero in the center of the mass distribution and large only outside the remote peripheric regions. Indeed, the DM density $\rho_D = Qg$ is expected to peak farther out, outside of the less massive front-and-end tip of the mass distribution, the tip where the stars that went through the collision center are located. However, the maximum of the cumulative DM $M_{<}$ is reached far outside the front-end tip P, or outside the rim of the mass distribution, and far behind the end tip P' (on the other side, on the side of the stars of the smaller of the colliding clusters that went through the collision center, just as observed). The DM density ρ_D decreases with the distance r

from the front point P (or end point P') slightly faster than $1/r^2$. *However, the cumulative $M_<$ increases strongly, almost proportional to r , peaking very far away out from the end points, as observed. This is similar to the seemingly unlimited increase of $M_<$ in halos of isolated galaxies, Eq. (4). For galaxy clusters the cumulative DM predicted by our key also increases proportionally to the distance r from P or r' from P', but slightly more slowly, with a density slightly below $1/r^2$, because the OM distribution is very elongated, and not spherically symmetric. This creates the impression of the DM outrunning OM at both ends.*

To also cover the even higher and highest scales S , with $S \gg L'$, of the OU above the "reference length" of about $L' \approx 10^9$ ly, for the density of DM, we could write

$$\rho_D = gQ(L'/S) \text{Log}(1+S/L'). \quad (25)$$

This extended DM key, applicable for any scale (or characteristic length S of the structure considered, including $S < L'$), can be used for possible "hyper-clusters of superclusters," if found, and for **all scales** $0 < S \leq R_u$ (where the additional factors in Eq. (25) are reduced to unity for $S \ll L'$). From Eq. (25) we see that it lowers the values of $Q = \lambda g'^2$ and ρ_D from the lower-scales amount in Eq. (14) for all higher scales $S > L'$. Eventually, the reference length L' could be fine-tuned, so the DM density (14) would add up to the total DM (about five time the OM) in the OU. Eq. (25) needs to be tested and verified.

Indeed, an important part of the universal DM will be accreted on the intergalactic ridges of gravitationally self-interacting aggregate of "DM plus intergalactic gas and dust," generating the branches of the cosmic web.

5. Dark Matter in the Observable Universe

This section presents an alternative derivation of DM, the initial derivation, based on EE and an electromagnetic analogy. In the first two sections here, it yields the rather simple, transparent and intuitive results for the percentage of directly generated DM in the OU.

Einstein's equations contain the metric tensor $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, defining the gravitational potential and resulting field, *on both sides*, on the l.h.s. in the Ricci tensor, and on the r. h. s. as part of its source, being present in $t_{\mu\nu}$, the gravitational field part of $T_{\mu\nu}$:

$$R_{\mu\nu} - g_{\mu\nu} \mathcal{R}/2 - \Lambda g_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}. \quad (26)$$

Here $R_{\mu\nu}$ is known as the Ricci tensor, obtained by contraction from the curvature tensor $R^\lambda_{\mu\nu\sigma}$, $\mathcal{R}$ is the curvature scalar obtained from R^μ_μ by further contraction, Λ the cosmological constant that describes the level of DE, $T_{\mu\nu}$ is the total energy-momentum-stress tensor, G Newton's constant of universal gravitation and c the speed of light. The field of the deviations $h_{\mu\nu}$, from the Euclidian metric (1, -1, -1, -1), contributes a gravitational field part, $t_{\mu\nu}$, to the total energy-momentum tensor $T_{\mu\nu}$. An

expression for $t_{\mu\nu}$, a restricted, or quasi-tensor, up to second order in $h_{\mu\nu}$, is given by Weinberg⁴ in Eq. 7.6.14. We used this expression for t_{00} earlier³ to calculate the total DM in energy units, W_{tot} in the OU analytically,³ to second order in $h_{\mu\nu}$ only, in spherical symmetry and in cylindrical flow symmetry. We found $W_{\text{tot}}=1.2 \text{ Gm}^2\text{N}^2/R_u$, in spherical symmetry, where R_u is the co-moving radius of the OU, and N is the number of galaxies of average mass m in the universe. By dividing W_{tot} with the rest energy of OM, Nmc^2 , we obtain the ratio of DM to OM in the OU

$$s'' = 1.2(4/3)\text{Gm}^2\text{N}'\text{N}/Nmc^2 = 0.8\text{N}'r_s. \quad (27)$$

This result, $0.8\text{N}'r_s$, is twice the classical result, derived³ with quasi-Newtonian theory. Here we introduced $\text{N}'=3\text{N}/4R_u$, the number of galaxies per unit length in the z -direction at the "equator" of the spherical model of the OU. We denoted by r_s the average Schwarzschild radius $r_s=2\text{Gm}/c^2$ of a galaxy, with m its average mass. When the universe is considered as a column moving with a certain hyperbolic angle to the time axis and (eventually virtual) spatial velocity $\mathbf{v}$ in the z -direction, we show below with an electrodynamic analogy, that the ratio of DM to OM should indeed be $s''=\text{N}'r_s$, as displayed in Eq. (27). Here N' is again the number of galaxies per unit length of the column.

To verify, we note that each galaxy moves along its world line. Consider the whole OU as the 3 D cross section of a

column moving along its world line or world corridor. Consider it also in 3 dimensions as a long column embedded in the rest of the universe, moving along its axis with a velocity vector $\mathbf{v}$ (eventually virtual), determined by the hyperbolic angle to the time axis.

Starting from an electrodynamic analogy to electric currents in a semiconductor or metallic wire, we show that, in the analog gravitational case, the DM ratio to OM in mass or energy is $\text{N}'r_s$, with the above notations. We mention this analogy here, both for historic reasons, and since it makes the notion of DM more accessible to a general readership.

Indeed, to verify this, the kinetic energy of the ordered motion of electrons with drift velocity v in a wire of radius a' carrying the current $J=\text{N}'ev$, with N' electrons per unit length, has two components: the coherent (many-body) magnetic field energy $\int B^2 d\tau/8\pi$, proportional to N'^2 in cylindrical symmetry, and the (incoherent) mechanical energy $\text{N}'mv^2/2$, proportional to N' . Their ratio is

$$\begin{aligned} s &= [\int B^2 d\tau/8\pi] / [\text{N}'mv^2/2] = \{ \int [4J^2/8\pi c^2] \\ &[2\pi r dr/r^2] \} / \{ \text{N}'mv^2/2 \} \\ &= e^2 \text{N}'^2 v^2 [0.25 + \ln(R_0/a')] / [c^2 \text{N}'mv^2/2] \\ &= 2\text{N}'(e^2/mc^2) [0.25 + \ln(R_0/a')] \approx 2\text{N}'r_e. \end{aligned} \quad (28)$$

Here $r_e=e^2/mc^2=2.8\cdot 10^{-13} \text{ cm}$ is the classical radius of the electron in CGS Gauss non-rationalized units. In Eqs. (28) and (29) the field energy *inside* the wire of radius a' and *inside* the radius a of the column, was included, and displayed explicitly both times in the last rectangular brackets through the

term 0.25. The logarithm describes a cutoff of the integrals by imagining an arbitrarily large closed circuit of radius R_0 . The expression in rectangular brackets, containing the logarithm, was approximated with the unity in the final result.

By replacing now $\mathbf{B}=\mathbf{v}\times\mathbf{E}/c$ with a quasi-Newtonian $\mathbf{v}\times\mathbf{g}/c$ for the kinetic gravitational field, we obtain³ the same way as $s=2N'r_e$ for the electrical case, *in the gravitational case* $s''=N'r_s$. This parameter s'' is again *independent of v*. It is the coherent (many-body) kinetic gravitational field energy of the galaxies, divided by the mechanical $N'mv^2/2$, both per unit length along the world corridor of radius $R_u=a$ of the OU in 3 dimensional space:

$$\begin{aligned} s'' &= s_g = [\int (\mathbf{v}\times\mathbf{g}/c)^2 d\tau / 8\pi G] / [N'mv^2/2] \\ &= [4Gm^2N'^2v^2 / (8\pi c^2 N'mv^2/2)] \int 2\pi r dr / r^2 \\ &= 2(GN'm/c^2)[0.25 + \ln(R_0/a)] \approx 2N'Gm/c^2 \\ &= N'r_s. \end{aligned} \quad (29)$$

Here $r_s=2Gm/c^2$ is again the average Schwarzschild radius of a galaxy.

Our Ansatz $\mathbf{v}\times\mathbf{g}/c$ in Eq. (29) was verified in Sec. 2.4 of our earlier paper³, based on an approximation of t_{uv} , derived from EE. A discussion of the electromagnetic analogy is given Weinberg⁴ pp. 161-171, and by Williams and Inan¹⁰.

In our analogy, the coherent kinetic energy part resulting from the magnetic field in a semiconductor or metallic wire corresponds to the coherent kinetic gravitational energy part of the universe, that manifests itself as its DM mass times $v^2/2$. Our detailed, laborious, calculations³ of this DM, both classically and in second-

order GR, both in cylindrical flow and in static spherical symmetry, yielded 80% of the factor 5, i.e., 4, as the ratio DM/OM=4. This has strengthened our confidence that a rigorous solution of EE would indeed yield 5 times more DM (as defined by us) than OM in the OU. It inspired the subsequent successful exact calculation of DM in all lower-scale structures of the universe with our DM key, shown here.

For example, if all galaxies would be like our Milky Way with $M\approx 10^{45}$ g, we would obtain

$$\begin{aligned} r_s(\text{galaxy}) &= 2GM/c^2 = 13.3 \cdot 10^{-8} \cdot 10^{45} / 9 \cdot 10^{20} \\ &= 1.48 \cdot 10^{17} \text{cm}. \end{aligned} \quad (30)$$

Thus, in the OU with co-moving radius $R=R_u=46.5 \cdot 10^9 \text{ly} = 4.4 \cdot 10^{28}$ cm, with about $N=2 \cdot 10^{12}$ galaxies, we can approximate the number of galaxies per unit length in cylindrical flow symmetry, by multiplying the average concentration of galaxies $3N/4\pi R_u^3$ with the equatorial area of the OU, πR_u^2 :

$$\begin{aligned} N' &= [2 \cdot 10^{12} / 1.33 \pi R_u^3] [\pi R_u^2] = 6 \cdot 10^{12} / 4 R_u \\ &= 1.5 \cdot 10^{12} / 4.4 \cdot 10^{28} \text{cm} = 3.4 \cdot 10^{-17} \text{cm}^{-1}. \end{aligned} \quad (31)$$

Therefore, we obtain

$$N'r_s = 3.4 \cdot 10^{-17} \text{cm}^{-1} \cdot 1.48 \cdot 10^{17} \text{cm} = \mathbf{5.032}. \quad (32)$$

The universe would thus contain about 5 times more DM as OM, in agreement with the present standard Λ CDM model, if our Milky Way could be considered an average galaxy. As we see below, it's above average, but DM becomes its own source to considerable extent from EE, and that seems to make up for it, correcting this low-order result. This result may seem to be

fortuitous, but the order of magnitude is important.

This interdisciplinary paper and the electromagnetic analogy used above were inspired by (the gravodynamic form of) our *quantum theory of the ubiquitous fundamental 1/f noise (Q1/fN)*, practically important in its QED form, in electrical engineering³ and in most of high-technology.^{11,12} Anticipating the modern theory of decoherence¹³ by decades, it used the "Einselected states," long before they were discovered. Due to this anticipation, Q1/fN was misunderstood by some influential theorists 10 years after its successful introduction, and was discredited erroneously. However, Q1/fN explains the fundamental, universal, 1/f noise and is expressed by simple universal engineering formulas for the new, "physical cross sections and process rates" of quantum mechanics, *and coherently, also for any (electrical or neutral-matter) current*. This is useful for the current US CHIPS program. Its "Einselected states¹³," or "pointer states" are more stable to the dephasing interactions with the environment. *Indeed, Eqs. (28) and (29) also yielded the ratio between coherent and conventional quantum 1/f effects in our gravodynamic quantum theory¹⁴ of 1/f noise.* Like the later-developed, well-known theory of decoherence (see, e.g.,¹³), which it anticipated since 1975, this quantum 1/f theory represents a new basic aspect of quantum mechanics (like decoherence), bridging the "quantum divide¹⁵" to our macro-world, even to astrophysics.

6. Dark Matter Generated in Lower-Scale Structures

In lower level structures the own, classically generated, Newtonian local gravitational field corresponds to a negligible energy or mass, compared with the OM. Indeed, even for a whole disk-shaped spiral galaxy, with about 10^{11} stars moving *along the axis of the galaxy* with disk-thickness $10^4 \text{ ly} = 9.5 \cdot 10^{21} \text{ cm}$, we obtain

$$N' = 10^{11} / 10^4 \text{ ly} = 10^{11} / 9.5 \cdot 10^{21} \text{ cm} \\ = 1.05 \cdot 10^{-11} \text{ cm}^{-1}. \quad (33)$$

Considering the average mass of a star $2 \cdot 10^{33} \text{ g}$, like the mass of the sun, with $r_s = 2GM_{\odot}/c^2 = 3 \text{ km}$, we obtain

$$N'r_s = 1.05 \cdot 10^{-11} \text{ cm}^{-1} \cdot 3 \cdot 10^5 \text{ cm} = 3.15 \cdot 10^{-6}. \quad (34)$$

This is a very small fraction of DM, which is generated at the level of a galaxy, compared to OM. If we consider (real or virtual) motion of the galaxy along a diameter of the disk, we obtain an even about 20 times smaller result. *In conclusion, the amount of DM generated in galaxies and in the even lower levels of the universe, such as planetary systems, is very small and can't explain the observed DM effects. Historically, this led to neglect of the energy of the gravitational field in galaxies and their clusters, in the presently favored LCDM model, introducing instead DM in the form of hypothetical hidden particles. The self-interaction of the gravitational field in EE results in the accretion of the large gravitational energy of the OU on the g of lower scale structures, such as galaxies or*

teir clusters. This removes the need for hidden DM particles.

7. Application of our DM Key to MOND.

The "Modified Newtonian Dynamics" (MOND) claims^{16,17,18,19} that Newton's law of universal gravitation, and implicitly GR, break down at very low accelerations, below 10^{-10} m/s^2 (corresponding to very large distances from a large central mass). According to the very successful MOND, the $1/r^2$ decrease of the gravitational acceleration $g_N = Gm/r^2$ at large distances r stops and needs to be replaced by something like a constant $a_0 \approx 1.2 \cdot 10^{-10} \text{ m/s}^2$, in fact by a geometric mean between the Newtonian gravitational field g_N and a_0 . *We show here that MOND and its partial successes can be understood, and derived, on the basis of our DM key.* We are thus relying on EE and its NA, to derive MOND, rather than demonstrating their breakdown.

According to the MOND algorithm, all very small Newtonian gravitational accelerations $g_N < 10^{-8} \text{ cm/s}^2$ need to be replaced by g from

$$g^2 = g_N a_0. \quad (35)$$

In general, g_N is replaced by *MOND*^{17,18,19} through $g = v(g_N/a_0)g_N$, where

$$v(x) \rightarrow 1 \text{ for } x > 1 \text{ and } v(x) \rightarrow x^{-1/2} \text{ for } x < 1. \quad (36)$$

We show here that the approximate solution of the EE and its NA expressed by our DM key validates the phenomenological MOND model to large extent. It explains the nature of DM and derives an analytical formula that defines a lowest acceleration a_0

for any galaxy, cluster, or other cosmic system of masses.

As we see from Eq. (6), for a point mass M (or "galaxy" with central mass geometrical size $a=0$), at a distance r , the centripetal acceleration g of a star in a galaxy is obtained with our key from

$$\begin{aligned} v^2/r &= (GM/r) \{1/r + 4\pi GQ\} = g \\ &= (GM/r) \{1/r + 1.164 \cdot 10^{-20} \text{ cm}^{-1}\}. \end{aligned} \quad (37)$$

By squaring Eq (37), we can write

$$g^2 = v^4/r^2 = (GM/r^2) \{GM/r^2 + (G^2M/r)8\pi Q + MG^3(4\pi Q)^2\}. \quad (38)$$

The last term in curly brackets, surviving at all larger distances r , yields *our remaining minimal acceleration* a'_0 . It does not decrease with r , and exactly corresponds to the definition in Eq. (35), with $g_N = GM/r^2$:

$$\begin{aligned} a'_0 &= MG^3(4\pi Q)^2 = 9 \cdot 10^{-52} \text{ (cm/s}^2\text{g)}M \\ &= 1.26 \cdot 10^{-8} \text{ cm/s}^2 = 1.26 \cdot 10^{-10} \text{ m/s}^2. \end{aligned} \quad (39)$$

The last two forms of Eq. (39) are again for a galactic central mass of $M_g = 1.4 \cdot 10^{43} \text{ g}$, corresponding to the average mass of a galaxy in the OU, Eq. (21). Our result for a'_0 is in general agreement with the $a_0 \approx 1.2 \cdot 10^{-10} \text{ m/s}^2$ introduced by Milgrom²⁰ and others²¹ for galaxies and clusters thereof. However, we notice that our smallest acceleration a'_0 is proportional to the central mass M . *Our universal constant* defined by Eq. (39) is

$$a'_0/M = G^3(4\pi Q)^2 = 9 \cdot 10^{-52} \text{ cm/s}^2\text{g}. \quad (40)$$

Another universal constant we introduce is the asymptotic squared velocity per unit central mass $v_o^2/M = g \cdot r/M$

$$v_o^2/M = 4\pi QG^2 = 7.76 \cdot 10^{-29} \text{ cm}^2/\text{gs}^2. \quad (41).$$

We note that $4\pi\lambda g'^2 G \equiv 4\pi QG = 1.164 \cdot 10^{-20} \text{ cm}^{-1} = 1/r_0$, where $r_0 = 9000 \text{ ly}$ is the universal distance at which the centripetal acceleration caused by DM gets to equal the one caused by OM, in the special case $a=0$, with M concentrated at $r=0$. This universal distance, independent of M is usually much smaller than the radius of the galaxy, so DM causes usually the centripetal acceleration, resulting in constant speed of the peripheric stars.

It is well known that MOND runs into difficulties at scales larger and lower than galactic. It does not work at scales below 1 light year, and struggles in clusters of galaxies.²¹ Our derived generalization of MOND is analytical and eliminates the troubling nonlinearities introduced by the non-analytical character of MOND in the presence of a small external acceleration, below a_0 . Such troubling, unphysical effects violating GR and its NA, appear even in the inner Solar System.²² All these problems are eliminated by our analytical generalization of MOND. Our derivation shows that no modification of Newton's or EE is needed in order to eliminate the need of hidden DM particle approaches in Λ CDM at any scale.

This concludes our derivation of the MOND phenomenological approximation and of its generalization for all scales, from EE.

8. Application of the DM Key to Wide Binaries

Consider a star of mass m_1 and radius a_1 . From Eq. (4') in the Main Text, applied

now for individual stars of mass m_1 and radius a_1 , the corresponding gravitational field is expected to be

$$g = m_1 G [(1 - 3\pi Q G a_1)/r^2 + 4\pi Q G / r]. \quad (42)$$

This corresponds to a potential

$$V_1 = -m_1 G [(1 - 3\pi Q G a_1)/r + 4\pi Q G \ln(R/r)], \quad (43)$$

Here R is a reference distance. The potential energy W of a system of two stars with masses m_1 and m_2 , of radii a_1 and a_2 is then, in a first approximation that assumes a superposition, $W = (m_2 V_1 + m_1 V_2)/2$,

$$\begin{aligned} W &= -G m_1 m_2 [(1 - 3\pi Q G a_1)/r_{12} + 4\pi Q G \ln(R/r_{12})] \\ &= -G \mu M [(1 - 3\pi Q G a_1)/r_{12} + 4\pi Q G \ln(R/r_{12})]. \end{aligned} \quad (44)$$

Here $\mathbf{r}_{12} = \mathbf{r}_2 - \mathbf{r}_1$ and $a = (a_1 + a_2)/2$. μ is the reduced mass $m_1 m_2 / (m_1 + m_2)$ of the two stars, M their total mass. For a binary considered by Chae²³, with $|r_{12}| < 100 \text{ pc} = 320 \text{ ly} = D'$, the relative error from superposing the accreted halo terms is small. It is approximated by the second term in rectangular brackets in Eq. (42) with $r = D'$, compared to $1/D'^2$

$$\begin{aligned} &4\pi G Q D' \\ &< 12.56 \cdot 6.67 \cdot 10^{-8} \cdot 1.39 \cdot 10^{-16} \cdot 320 \cdot 9.5 \cdot 10^{17} \\ &= 0.071 < 1. \end{aligned} \quad (45)$$

We mention that an estimate of the average inter-stellar distance in a spiral galaxy is of the order of 10 ly, limiting the usual size of stellar halos. Our DM key, combined with our result $\rho_\Lambda = 5.8 \cdot 10^{-30} \text{ g/cm}^3$ in Sec. 3 MT, shows that even completely isolated stellar halos of stars with size like the sun could not extend beyond $D = 100 \text{ ly}$, because they would drown in DE. At $r \approx D$

the cumulative mass $M_{<}$ of the Halo is completely compensated by the negative DE. The latter grows proportional to r^3 , while $M_{<}$ is proportional to r . We thus will set $R \approx D = 100$ ly in the logarithms. For isolated typical galaxy, this distance would be about $D' \approx 3 \cdot 10^7$ ly, much larger than half the distance from our Milky Way to Andromeda, mentioned after Eq. (6) to be about $1.25 \cdot 10^6$ ly.

For wide binaries considered by Chae²³ the effective gravitational potential in Eq. (44) including both OM and DM allows, *in a first approximation*, for integration of the two-body problem for wide star-binaries with $a \approx 0$. Indeed, in the center of mass coordinate system (CM), from

$$\mu d^2 \mathbf{r} / dt^2 = -G\mu M (1/r^2 + 4\pi GQ/r) \mathbf{r} / r, \quad (46)$$

where μ is the reduced mass $m_1 m_2 / (m_1 + m_2)$ of the two stars, M their total mass. In Eq. (46) we introduce the notation

$$(r_0)^{-1} = 4\pi GQ = 1.164 \cdot 10^{-20} \text{cm}^{-1} = A. \quad (47)$$

In Eq. (44) we have a central force, which leads to a potential energy,

$$V = -G\mu M [1/r + A \ln(R/r)]. \quad (48)$$

R corresponds to an additive constant in the energy. Since the force is central, the total angular momentum $\mathbf{L}$ is a prime integral, and the equation of the orbit for motion in a plane perpendicular to $\mathbf{L}$ is:

$$\mathbf{L} = \mu r^2 d\phi / dt = \text{Const.}; \quad d\phi / dt = \mathbf{L} / \mu r^2 \quad (49)$$

in plane polar coordinates. The Lagrangian or Hamilton-Jacobi formalism, e. g., can be used, in polar coordinates, to obtain with Eq. (49) the shape of the orbit

$$\phi = \int^r \mathbf{L} \cdot \{E + G\mu M [1/r + A \ln(R/r)] - \mathbf{L}^2 / 2\mu r^2\}^{-1/2} \cdot (dr / 2^{1/2} \mu^{1/2} r^2). \quad (50)$$

If $\ln(r/R) \approx \langle \ln(r/R) \rangle$ is considered constant *in a second approximation*, for not too large eccentricity of the orbit, one can integrate through arc cos and obtains the known elliptic orbit in polar coordinates in the center of mass system,

$$r = (\mathbf{L}^2 / \mu^2 GM) (1 + \epsilon \cos \phi)^{-1}. \quad (51)$$

Numerical integration can remove this approximation. Here $\mathbf{L}$ is the total internal orbital angular momentum of the binary system, and the eccentricity is given by

$$\epsilon = \{1 + 2[E + G\mu M A \langle \ln(R/r) \rangle] [\mathbf{L}^2 / \mu (\mu GM)^2]\}^{1/2}. \quad (52)$$

This shows that the stationary orbit is a slightly distorted ellipse.

We may examine the time averaged motion in this stationary state with the virial theorem. The latter is derived by assuming that the second derivative of the system's moment of inertia (of the two-star system) has a time average which is zero. In our case, in the CM, in time average, from Eq. (9)

$$2\langle K \rangle = \langle \mathbf{r} dV / d\mathbf{r} \rangle \text{ (virial theorem), i.e.,} \\ \langle \mu v^2 \rangle = \langle GM\mu / r \rangle + AG\mu M. \quad (53)$$

Here $\langle K \rangle$ and $\langle V \rangle$ are the average kinetic and potential energies. Comparing with Eq. (44), we see that the logarithm averages to unity here, because the average of $3\pi GQa / r_{12}$ drops out, for $a \ll r_{12}$. This supports the picture of a deformed elliptical motion, which could be better obtained by numerical integration in Eq. (50). The result should

match the very detailed measurements *and* improve the statistical results of Chae.²³

9. Conclusions

We started by calculating the energy of the gravitational field of the OU in four ways.³

Twice we calculated classically and twice by calculating the metric in GR,³ Both classically and from EE, we calculated³ in the spherically symmetric picture and in the cylindrically symmetric picture. In all cases the result coincided in order of magnitude with the rest energy mc^2 of the ordinary matter (OM) in the OU.

The relativistic calculation to second order in the deviations $h_{\mu\nu}$ from the Euclidian metric, yielded twice the best classical result, about 2 times the rest energy of the OM in the OU. We then assumed that a calculation to all order of $h_{\mu\nu}$ may indeed yield about 5 times the rest energy of the OM in the OU, as considered in the Λ CDM model. This could happen because the metric appears on both sides of the nonlinear EE. It may well self-generate itself at these large values of DM/OM, becoming in part its own source.

An independent electromagnetic analogy showed that the DM fraction associated with a stream of N' OM pieces per length unit along the stream, each of mass m , is $s''=2N'rs$. This is shown in Secs. 5 and 6 above, with rs as the Schwarzschild radius of one piece of mass m . Considering the OU as a stream of galaxies similar to the Milky Way, we initially obtained in Eq. (32)

the direct result $N'rs=5.032$, as the ratio between the gravitational field energy of OM and the rest energy of (baryonic) OM in the OU. This is also the ratio between the corresponding masses, between the DM and OM. It agrees again with Λ CDM, but without hidden DM particles, rather as a feature of space-time. Indeed, all these results led us to identify the energy of the huge coherent gravitational field of the OU as the mysterious DM.

However, Sec. 6 showed that the fraction of gravitational energy, i.e., DM, generated in a single galaxy from $s''=N'rs$ is about 10^{-6} , i.e., utterly negligible. This led in Sec. 2 above to investigate the way in which the large fraction of DM in the OU gets to be distributed over the lower-scale structures of the OU. The self-interaction of the gravitational field and energy at various levels of the OU, as implied by EE, eventually led to the straight derivation of our universal DM key in Sec. 2.

Our approach culminating with our "DM key" $\rho_D=Qg$, with $Q=1.39 \cdot 10^{-16} \text{ gs}^2/\text{cm}^4$ is validated both by its derivation, essentially from EE, and from its Newtonian Approximation (NA), *as well as by its fascinating results:*

1. Exact derivation of DM in any point of the OU, where the classical, Newtonian gravitational field g can be easily found by ignoring all DM and DE effects;

2. Easy analytical derivation of the galactic rotation curves of stars, gas and dust at all distances from the galactic center, very close and arbitrarily far, experimentally

verified, with the possible inclusion of the outer vicinity of the central black hole;

3. Elimination of the central spike predicted in simulations with DM hypothetical particles. The DM concentration increases from zero in the center of a galaxy proportional to r for $r < a$, as mentioned right before Eq. (4). This is the natural solution of the "Cuspy Halo Problem" present in all simulations with cold, warm, fuzzy, self-interacting, meta-cold, etc., DM²⁴;

4. First time easy analytical derivation of the DM concentration in the galactic halo at all distances from the galactic center, both very close and very far, beyond the prediction of all known models, in perfect agreement with the startling measurements of Misteale et al.⁵ They verified our $\rho_D \sim 1/r^2$, our cumulative $M < \sim r$ indirectly by gravitational lensing. They verified the rotation speed $v = \text{const}$ directly, from the Doppler effect in the 21 cm spectral line in the rotating Hydrogen. Their results showed no decrease in circular velocity v for isolated galaxies, up to distances of 1 Mpc = 3,261,564 ly, well beyond the predictions of all known DM models, in perfect agreement with our theory;

5. The analytical derivation of the Tully-Fisher relation for spiral galaxies;

6. The exact analytical derivation³ of the very small amounts of DM in the Solar System and any planetary system. The exact derivation of 134 grams of DM in any km³ at the earth's surface from $g = 980 \text{ cm/s}^2$.

Easily scaled down for any room on the earth;

7. The easy analytical derivation and generalization of the well-known MOND approximation algorithm, improving its general agreement in low-level structures of any scale. The definition of several related new universal constants was given. All this was derived in fact from EE and its NA, without any modifications;

8. Exact derivation of the Dark Energy concentration in any point of the OU. Exact derivation of the cosmological constant Λ , in final analysis again, only from EE and its NA, as a feature of space-time;

9. Derivation of the simple analytical expression $s = N'r_s$ for DM/OM from EE and its NA in Sec. 5. Based on an electromagnetic analogy,^{4,10} this simple relation, like $s = N'r_e$ in electrodynamics, considers a current of objects of mass m in gravodynamics, or of electrons in electrodynamics. It yields the classical and GR coherent energy fraction or the DM fraction.. It gives the ratio s of the coherent, electromagnetic kinetic energy to the incoherent kinetic energy of the electrons' drift motion. In the same way, it yields for a stream of objects of mass m the "DM divided by OM" ratio s without considering the "ensuing" accretion of coherent DM from the OU on the gravitational field g generated by lower level structures. "Ensuing" is in quotation marks, since it happens at the same time in fact. It is subsequent only in our mind, in our analysis of this nonlinear process, orchestrated by

EE, which is considered "multi-stage" only in our analysis, similar to a series expansion;

10. Most recent gravitational lensing observations by the James Webb Space Telescope (JWST) have shown that dark matter in the small galaxy clusters within the El Gordo supercluster is distributed more uniformly than previously thought, based on Λ CDM. This understanding emerges from a comprehensive analysis of JWST images of El Gordo, revealing distorted background galaxies that, when analyzed, indicate a smoother dark matter distribution within those smaller sub-structures compared to the lumpiness expected from Λ CDM simulations. We consider this smoother distribution as an indication that the classical Newtonian g (obtained without DM or DE) directs the DM accretion Q_g of OU-generated DM. This Newtonian g is maximal outside the small galaxy clusters, zero or minimal in their center, and therefore smoother;

11. The epistemological derivation of *the true nature of DM and DE*, was performed in an Appendix (or supplement) of this paper for the first time from our new vantage viewpoint, based on the new landscape that became visible through the present paper and reference 3. This Appendix also shows how and in what sense, DM, DE and Λ even exist, or do not

exist. This most exciting part, however, is not yet submitted for publication at the present time, unless requested by the Editors, since it is not quite free from informed conjectures.

In conclusion, in the course of our investigation, we have found various results actually based on EE solutions for a current of objects of mass m , for the whole OU, and for the various lower-level and higher-level structures of the OU in interaction. Some of these results came from the NA, partially applicable to the low gravitational fields found outside of black holes and neutron stars, some from the Schwarzschild solution, some from the large-scale gravitational field acting as its own source in EE, or from the field's self-interaction.

While the formal solution of EE for all the interacting levels and structures of the OU does not seem accessible for the present time, our partial results obtained here may be a reasonable approximation for the present time, an approach of surprising simplicity and accessibility for our readers.

ACKNOWLEDGMENT

We gladly acknowledge useful discussions with C. D. Frederick, P. F. Handel, T. F. George and E. L. Gibb.

APPENDIX #1

Supplement to the Paper "Derivation of Dark Matter, Dark Energy and MOND, as Features of Space Time"

The Nature of Dark Energy and Dark Matter

P. H. Handel^{1*} and K. E. Splett²

¹Department of Physics and Astronomy, University of Missouri-St. Louis & Center for Nanoscience; St. Louis MO 63121, USA.

²University of Missouri-St. Louis; present addr.: Petersroda, Sandersdorf-Brehna, Germany.

*Corresponding auth: handel@umsl.edu, phasenoisehandel@gmail.com, erikasplett@yahoo.de; use both!

I. The Geometric Aspect and Nature of Dark Matter and Dark Energy

The Nature of Dark Matter

We may now interpret what our paper has shown, in a different way. Dark matter (DM) and Dark Energy (DE) were derived for lower-scale structures according to general relativity (GR) in their local, over-all Euclidian or Newtonian, frame (E-frame) of reference (for an observer located in the center of the structure). DM was derived *as the gravitational energy of the observable universe (OU)*, accreted as DM on the local Newtonian gravitational field g , with a concentration

$$\rho_D = Qg \tag{1}$$

in each point/event. This allowed also the derivation of the baryonic Tully-Fisher relations for spiral galaxies, among other results. For all lower-scale structures at and below the scale of large super-clusters of galaxies, Q was fixed by the measured coefficient of these Tully-Fisher relations (averaged over a large number of different galaxies), at $Q = 1.39 \cdot 10^{-16} \text{ gs}^2/\text{cm}^4$. This yields negligible effects on planets, on planetary systems³, but not in galaxies, clusters and super clusters. There the DM obtained from Eq. (1) as a feature of space-time is larger, exactly describing through its gravitation, the rotation curves of stars, the galactic halos and the stability of clusters of galaxies. On all these lower scale structures of the OU, the cumulative DM contributions increase linearly with the distance r from an observer placed in the center of mass, e.g., from the galactic center, and ultimately also with the size of the structure (e.g., the galaxy). ***In this Supplement containing also informed conjectures, we will develop an alternative geometric-physical description of this accretion of DM from the OU on the local field g of lower scale structures. It will finally clarify the nature of DM and DE, by focusing on the local frame of reference. The conjectures should be verified before they are accepted as facts.***

We consider in Fig. 1 an analogy with a curved surface, e.g., a sphere or a portion of a spheric²⁵ or hyperbolic surface, and a plane tangent to it in a point P . In the vicinity of P , the spherical surface segment can be identified with its tangent plane. However, this local approximation with a plane also requires increasing corrections, as the area around P considered on the plane and the distance from the point of contact, increase to P' or P'' in Fig. 1.

The same way, in GR the local vicinity of a material event-point P in free fall is approximated by Newtonian physics as an Euclidian (E) "tangent" space, with no curvature, *the local "E-frame" of reference*. This Newtonian approximation (NA) requires increasing corrections as the size of the structure described in the local Newtonian E-frame approximation increases, e.g., to galactic dimensions. *We suggest that these corrections come in form of added DM, needed in order to continue describing, e. g., the rotation curves of peripheric stars on the E-plane at P' , P'' , with Newton's laws of motion, valid in the local E-frame, as we have seen in the main text.* These peripheric stars p' or p'' in the actual curved universe are also seen in the E-frame; they are perceived at P' and P'' on the light-cone of the E-frame tangent in observer's freely moving material event-point.

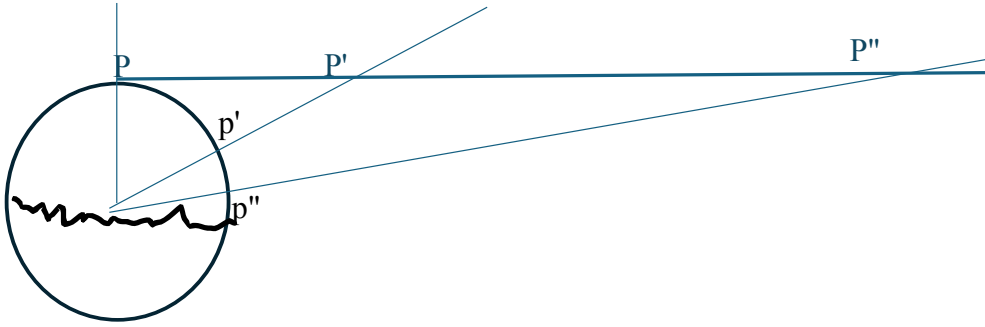


Fig. 1. A spherical polar segment (and the whole sphere), contacted in P by a plane that symbolizes the 4D Euclidian local frame space, with indefinite metric, the space of our perception.

The concentration of this added DM accretion can again be described as a function $f(g, g'')$, of the local gravitational field g and of a parameter g'' symbolizing the curvature of the OU in the point of contact, or event, P , e.g., at the center of a galaxy. We would then repeat the derivation that led to Eq. (2) **MT, (main text)** arriving to a similarity between:

- the accretion of universal DM on the local g , described in MT on one hand, and
- DM introduced as compensation for the deviation caused by the use of the local Newtonian E-frame to describe the actual curved universe, on the other hand.

In this local frame representation, the accreted concentration of DM could be given by a series of terms starting with $\rho'_D = Q'g$. Indeed, symbolically, the local g would be interacting with a g'' describing local universe's curvature effects, and we could expand the unknown correction function $f(g, g'')$ again in a Taylor series. The complete similarity means $g' = g''$ and $f = f'$, i.e., the present notions and notations correspond to the ones we introduced before. The linear increase of the accumulated DM halo, $M_<$ with distance r from the contact point P (e.g., from the center of the galaxy), would then correspond to the indefinitely increasing distance r in the local E-frame. It would also correspond to the unlimited increasing difference between the local frame and the actual curved, universe, with increasing extension. *We could thus understand the unlimited*

increase of the cumulative DM of an isolated galaxy with the distance r from the center, shown by Eq. (4) MT. This increase is zero at the contact point P, from Eq. (5) MT, eliminating the presence of a constant term $\rho_D(0)$ in the series expansion Eq. (1) MT and Eq. (1) here, with $\rho_D(0)=0$, and keeping only the first term, $Qg \equiv \lambda g g'^2$. This increase ends where the gravitational field switches direction to the neighboring galaxy, or structure in general. There, the description of the next structure of the universe begins with the local E-frame located in the center of this neighboring structure (because it is from *its* center, where we start applying Newton's laws of motion again).

The Nature of Dark Energy

In practice, in Astrophysics, also in the main text, the partial amounts of accreted DM, seen in our perceived Euclidian universe mainly through lensing effects, *are defined in our local E-frame* for various structures of the OU. *Since the space of our perception is Euclidian, DE is introduced in it to compensate the curving effects of the perceived DM. If all the DM is an effect caused by the automatic use of the extended local E-frame perception by humans, the compensating DE, required for the OU to be flat, is also an effect caused by the use of the extended local E-frame of our perception, defined by GR. Indeed, we are in free fall, with the whole solar system. This defines our Newtonian local E-frame, in fact our extended local E-frame, which we use to describe the whole perceived OU.* In this, our E-frame, a DE per volume unit, $\rho_\Lambda = 5.96 \cdot 10^{-30} \text{ g/cm}^3$ is shown in Eq. 18 of MT to be needed to keep the Newtonian E-frame of our human perception Euclidian, i.e., from becoming curved from the combined gravitational effects of the ordinary matter (OM) and DM. *This, in turn, explains why an accelerated Hubble expansion is observed in our perceived OU.*

We can see this compensation in the local E-frame of the OU, displayed in the local extended E-frame in Eq. (18) MT in a reasonable approximation:

$$4\pi G Q M_b L_g - \rho_\Lambda 4\pi L^3 / 3 = 0. \quad (2)$$

The first term on the left is the resulting galactic dark matter, perceived in the OU, and the second is the galactic compensating dark energy. This yields our predicted values of $\rho_\Lambda = 5.8 \cdot 10^{-30} \text{ g/cm}^3$ and $\Lambda = 1.08 \cdot 10^{-56} \text{ cm}^{-2}$, all fixed by the measured coefficient in the Tully-Fisher relation. *This discussion can explain the DM in our local E-frame of the perceived OU. It can at the same time explain the nature of DE necessary to also compensate the DM in the local E-frame. (Note: For a better approximation, in which OM is not neglected, we could attempt to replace Q by $1.2 Q$ in Eq. 18 MT.)*

The Actual Curved Universe

We have thus eliminated both, the local frame concepts of *DM and DE*, from the actual universe that exhibits a 4-D curvature, but keep them as observed, real, useful, concepts,

observed and defined in our local Newtonian E-frame of reference of GR. This explains in which sense they exist.

Our calculation in Eq. (18) MT, and in the whole Sec. 4 MT, was done in our local extended E-frame of the whole OU. It considered in Sec. 4 MT first the DM, resulting from all galaxies, each taken like the average galaxy, with an appropriate part of the space corresponding to the voids in the whole OU. As we mentioned, the real need for DM and DE is only perceived in our extended *local Newtonian E-frame*, for lower-scale structures, and for the whole OU: an Extended E-frame that we are always using unwittingly. The lower-scale structure of the part of the actual universe, corresponding to the OU, exhibits a hilly 4-D landscape with many kinds of curvature. In a general hierarchic understanding of the whole universe, the OU may correspond to a minuscule part of the universe, a small part that we perceive as flat in our local E-frame. It does not allow us at this point, to observe directly any possible global curvature of the whole universe, or absence thereof. *The hierarchy noticed in the lower-scale structures of the OU is observed to be limited at larger OU scales, by a reduction in the inhomogeneities and in the anisotropy, in the perceived, so called, "end of greatness."* The latter is an observed tendency to homogeneity and local isotropy at scales above 100 Mpc or $3.2 \cdot 10^8$ ly, in the OU, as perceived in our local E-frame, agreeing with the cosmological principle. *"End of greatness" could be an indication that the part of the actual universe, corresponding to our OU could also be more uniform, within the confines of this region.* In this picture, we have been perceiving the OU in our local, Euclidian frame, which is analogue in 4-D, to a 2-D tangent plane. A solution of Einstein's equation (EE) in the OU (or in a significant part of it) or in the whole curved universe at all scales, also containing our OU, is not known. We only have been able to "anticipate" some of the partial results of EE for the OU.

We can attempt to visualize this 4-D picture with a very familiar example: *displaying the spherical surface of a globus, or of the earth, on a wall map.* On this flat map of the earth, there are large distortions. The poles become polar lines. The polar regions are displayed greatly enlarged, compared to the equatorial regions. This could be analogous to the presence of about 14 times more DE than OM in our local E-frame, in the OU. On the other hand, two points on a very steep 60° slope, separated by 100 m would appear being separated only by 50 m horizontal distance on a very detailed map (perhaps less, if cosh applies instead of cos). This could correspond to the DM distortions noticed as lensing effects in our E-frame, looking at parts of the OU, displaying corresponding parts of the curved universe. These 4D curvatures (responsible for $DM \approx 5$ times the OM) are much larger than the corresponding altitude differences on the earth's surface in this simple analogy. In principle, however, the map distortions would correspond in this analogy to DM and DE seen in our 4D perceived Euclidian OU.

Note that the shortest distance lines (geodesics) connections, followed, e.g., by commercial airlines, appear curved to the North in the Northern hemisphere on the flat world map. This would be interpreted as DM and DE-caused gravitational effects by intelligent beings living in such a flat 2-D world. (Could an improved reference frame yielding something like a Mercator projection be possible, re-defining DM and DE?)

This oversimplified analogy with a globus is introduced here for a visualization, while the OU is only a part of the actual curved universe.

Observations

Experimentally, we note that quite different amounts of DM are observed in different galaxies. The observed ratio between the halo mass M_H and the stellar mass M_S of a galaxy^{26,27} is a smooth function $f(\xi)$ of $\xi = M_H/M_S$, with a minimum of about 30, about where the Milky Way galaxy is. The function increases both toward slightly larger ratios ξ (approximating DM/OM) and toward lower ratios. It has a scatter by a factor of less than two for large galaxies, but the scatter in the ξ ratio is much larger for small galaxies. For instance, the ultra-diffuse galaxy NGC1052–DF2 has no detectable amounts of dark matter²⁸, $\xi \approx 0$.

Indeed, we expect larger galaxies to form on branches of the cosmic web, which are ridges of DM and hydrogen gas filaments. They have a larger ξ , because of the bunching tendency of the gravitational field energy, i.e., of DM, over cosmic distances, which also created the cosmic web. Indeed, the bunching that led to the web is caused by the self-interaction of the gravitational field that always carries mass. This self-interaction with more abundant gas and dust could also cause the increased ξ in the large galaxies found on branches of the web.

The isolated galaxies located in voids of the cosmic web, like our Milky Way, do not benefit from this augmentation of ξ , seen on branches of the web, and are expected therefore, to show somewhat lower ξ values, e.g., 30. On the other hand, some galaxies may be located in places close to a curvature reversion (inflexion) region or a saddle-point in the curvature landscape of the many maxima, or bumps, present locally in that cosmic web region of the universe, as perceived in our extended E-frame. Therefore, they show smaller ξ , i.e, lower amounts of DM, with strongly varying ξ values, depending on their exact position.

For example, consider a galaxy located in the center of a void separating two huge clusters of galaxies of similar size, right at the center of mass. The gravitational field caused by each cluster is cancelled by the opposite cluster yielding $g=0$. That galaxy²⁸ will show very little DM, according to our DM key.

In conclusion, our flat universe may only be the flat image of the actual, curved universe, in the Newtonian E-frame of our E-frame perception of the actual universe. The discussion presented below in Sec. II below (Negative mass aspect) thus remains applicable as the "internal dynamics" of this flat perceived image of the universe, explaining the classical relation between the apparent DM and DE in this flat space of our present human astronomic perception.

Discussion.

It is fair to mention that in a way, the geometrically defined DM and DE exist and still do not exist at the same time. Indeed, as we have seen, DM and DE certainly are observed as realities in the local, Newtonian E-frame of reference of our perception. Nevertheless, they may not exist in the actual OU, which exists objectively, and which we are a part of.

The formation of the cosmic web can be understood both ways. In our extended local E-frame, intergalactic gas, galaxies and eventually dust, self-interact gravitationally; more universal DM accretes on their common gravitational field and also self-interacts; and they all interact among themselves gravitationally creating the cosmic web, decorated by galaxies and clusters in the OU.

In our geometric interpretation, intergalactic gas, galaxies and eventually dust self-interact gravitationally, creating the cosmic web in the actual universe and curving it according to EE in curvature ridges or branches, without DM and DE. In the extended E-frame of our perception, this is observed along with the DM shaped according to these ridges and along with the accompanying universal DE contributions. This way we obtain the same cosmic web, decorated with galaxies and clusters thereof in our E-frame.

II. The Negative Mass Aspect of Dark Matter and Dark Energy

1. The main text has used our previous paper³, which explained how the large amount of gravitational field energy generated according to Einstein's field Equation (EE) of general relativity (GR) acts: it acts mainly as its own source, is about 5 times larger than the total amount of ordinary (mainly Baryonic) matter, and should be interpreted as universal Dark Matter;

2. On this basis, we now found that *this large universal DM* manifests itself, being *accreted on the classical Newtonian gravitational field g of lower-scale structures* (planetary systems, galaxies, their clusters and superclusters), with a density $\rho_D = \lambda g g'^2 = Qg$. Here $Q = \lambda g'^2$ is the accreted DM concentration per unit Newtonian field g . It is fixed along with the derivation of the Tully-Fisher relations for spiral galaxies;

3. The calculated accreted DM turns out to be negligible in planetary systems, considerable in binary star systems, and dominant in galaxies, clusters and superclusters. It allows the

derivation of the rotation curves and of DM halos in galaxies, without any central halo spike, in very good agreement with observation;

4. Based on the cosmological principle, we also showed in this Manuscript how, by assuming a flat universe of our perception, our mentioned DM density formula also allows for the derivation of the cosmological constant Λ and of the dark energy (DE) density, in very good agreement with the observations.

5. *This exact derivation of Λ suggests a new physical understanding of the DE, which is now first explained in a qualitative way here in the spirit of Einstein's field Equation (EE). The equivalent geometric aspect of DM and DE was explained in the previous section, revealing their nature.*

Indeed, for an observer located far outside the (e.g., galactic) system, the internal gravitational binding energy W of a lower-scale structure or system (including that resulting from all the accreted DM!) considerably reduces its mass by W/c^2 and thus lowers the Newtonian attraction force on a mass M at a distance D by GMW/c^2D^2 , where D is the effective distance to the mass M . Based on our interpretation of DM, this corresponds to a gravitational repulsion between DM on one side, and external ordinary matter, located outside the lower-scale system considered. ***For the world outside the system considered, it looks as if DM, and in general, the gravitational field energy, would have negative mass. This is how it will be seen from outside.***

A similar repulsive force contribution will arise between two galaxies. This time, however, the repulsive contribution from the binding energy of the accreted DM density $Qg \equiv \lambda g g'^2$ may not be negligible. For any test mass in the inter-galactic space, this repulsive force, from the internal negative binding energy of the galaxy, will be felt along with the attractive force caused by ordinary matter and is likely to prevail. This prevailing repulsive force field is present in any point of the inter-galactic space and is cumulated in coherent superpositions. It originates from all gravitational binding energies, including binding energies present in galaxy clusters or superclusters, including binding energies between these, all being hugely augmented by the accreted density $Qg \equiv \lambda g g'^2$ of universal DM. This proliferation of repulsive DM binding energies (that become in majority their own source in the energy-momentum tensor on the r.h.s. of EE), causes the flattening of the universe and the accelerated expansion of the universe. The latter is caused by over-compensation of the positive sources in the energy-momentum tensor on the r.h.s. of EE. It also gives the free space the negative energy density, and, if not included explicitly in $T_{\mu\nu}$, it is displayed in EE by the cosmological constant Λ . For the universe to be flat, it will have to satisfy the relations derived in Sec. 3 MT. However, these considerations suggest an additional source of DE.

Indeed, to this point we have neglected the largest concentration of negative gravitational binding energy introduced by GR. This is found in black holes. Thus, instead of considering only the binding energy defined by the internal gravitational field of a galaxy and the inter-galactic

field as the source of the curvature-reducing DE, one should also include the contribution of the super-massive black hole in the galactic center and of other galactic black holes, of stellar origin.

At this point we mention the recent efforts²⁹ to identify black holes as the source of DE. We identify here all gravitational field binding energies in the universe as the source of DE. Thus we consider all binding energies, including also the smaller contribution from black holes, as the source of DE. The DE is therefore also effectively defined by the EE, as a physical feature of our perceived space-time only.

This way it is possible to physically understand the nature of "dark energy" on the basis of GR, as a feature of space-time in our perceived universe, determined by Λ . This was in essence described, in other terms, already by Friedmann³⁰, who was unaware of the accelerated expansion of the universe.

The question arises, how can EE describe both the attractive, positive mass role of DM, and the repulsive, negative role at the same time? This contradictory reality is described in EE only through the cosmological constant Λ , which can be introduced³¹ as an integration constant. On the other hand, in the previous section we found a more fundamental description of DM and DE as compensating distortions, *that make the local Newtonian E-frame of our perception of the OU flat*, as required by GR, with $\Omega_k=0$ without any observed curvature. This could help easier reconcile the mentioned formal contradictory reality in EE.

SCIENCE AUDIO CAST: Klick, or paste in the link below to listen to an explained Audio Version of this paper, with its Appendix 1: (then click on Audio Overview, and then on the small triangle, to listen to the audio cast):

Appeal to Publish: A Complete Solution to Dark Matter and Dark Energy

<https://notebooklm.google.com/notebook/a4edb930-39d3-47d2-bba3-444e29502169>

References

¹. M. Tanabashi et. al. *Astrophysical Constants and Parameters*. *Physical Review D. Particle Data Group*. **98**, 030001 (2019); *doi:10.1103/PhysRevD.98.030001*.

². Planck 2018 results. VI. Cosmological parameters. arXiv>[astro-ph](https://arxiv.org/abs/1807.06209)>arXiv:1807.06209.

³. P. H. Handel and K. E. Splett, Calculation of Dark Matter as a feature of Space-Time, Springer Nature "Foundations of Physics," **53** (5) 2023, DOI 10.1007/s10701-023-00705-x.

⁴. S. Weinberg, *Gravitation and Cosmology*. Wiley and Sons, 1972.

5. T. Mistele, S. McGaugh, F. Lelli, J. Scombert and P. Li, Indefinitely Flat Circular Velocities and the Baryonic Tully-Fisher Relation from Weak Lensing, arXiv:2406.09685v1 [astro-ph.GA] 14 Jun 2024
6. A. Deur, Implications of Graviton-Graviton Interactions on Dark Matter. *Physics Lett. B* **676**, 21-24 (2009).
7. J. Trachternach, J. et al. The baryonic Tully-Fisher relation and its implication for dark matter halos. *Astronomy and Astrophysics*, **505**, 577-587 (2009).
8. Yiping Wang et. al., Robust Quantum Control via Multipath Interference for Thousandfold Phase Amplification in a Resonant Atom Interferometer. *Phys. Rev. Lett.* **133**, 243403
9. E. M. Silich *et al.*, ICM-SHOX. I. Methodology Overview and Discovery of a Gas–Dark Matter Velocity Decoupling in the MACS J0018.5 +1626 Merger, *Ap. J.* **968**, 74 (2024) DOI 10.3847/1538-4357/ad3fb5e.

10. L. L. Williams and N. Inan, Maxwellian Mirages in General Relativity. *New J. Phys.* **23**, 053019 (2021).
11. A. van der Ziel, Unified Presentation of 1/f Noise in Electronic Devices; Fundamental 1/f Noise Sources, *Proc. of the IEEE*, **76**, 233-258 (1988).
12. P. H. Handel and A. G. Tournier, Nanoscale Engineering for Reducing Phase Noise in Electronic Devices. *Proc. of the IEEE* **93**, 1784-1814 (2005).
13. W.H. Zurek: Decoherence, Einselection and the Existential Interpretation (the rough guide), *Phil. Trans. R. Soc. Lond. A* (1998) **356**, pp. 1793-1821.
14. P.H. Handel: "Coherent and Conventional Gravidynamic Quantum 1/f Effects," *Proc. Intl. Conf. on Noise and Fluctuations (ICNF 2019 Neuchatel)* Eds. Ch. Enz and F. Pascal, Imprimerie EPFL Lausanne, pp. 140-143.
15. T. Folger, *Scientific American* July 2018, p. 30-35
16. M. A. Milgrom, Modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophysical Journal* **270**, 365–370 (1983)
17. M. Milgrom, A modification of the Newtonian dynamics - *Implications for galaxies*. *Astrophys. J.* **270**, 371–383 (1983)
18. M. Milgrom, A modification of the Newtonian dynamics - *Implications for galaxy systems*. *Astrophys. J.* **270**, 384 (1983).

19. J. Beckenstein and M. Milgrom, Does the missing mass problem signal the breakdown of Newtonian gravity? *Astrophys. J.* **286**, 7-14 (1984).
20. Milgrom, M., Models of modified-inertia formulation of MOND, *Phys. Rev. D*, 106, 064060 (2022).
21. S. Mazurenko, I. Banik and P. Kroupa, A simultaneous solution to the Hubble tension and observed bulk flow within $250 h^{-1}$ Mpc, *MNRAS* **527**, 4388–4396 (2024) <https://doi.org/10.1093/mnras/stad3357>.
22. M. Milgrom, MOND effects in the inner Solar system, *Mon. Not. R. Astron. Soc.* 399, 474–486 (2009); doi:10.1111/j.1365-2966.2009.15302.x.
23. Kyu-Hyun Chae, Breakdown of the Newton–Einstein standard gravity at low Acceleration in internal dynamics of wide binary stars *Astrophys. J.*, 952:128, (33 pp) (2023).
24. McGaugh, S.S. et. al. The rotation velocity attributable to dark matter at intermediate radii in disk galaxies., *The Astrophysical Journal*. **659** (1): 149–161 (2007). [arXiv: astro-ph/0612410](https://arxiv.org/abs/astro-ph/0612410). [Bibcode: 2007ApJ...659..149M](https://ui.adsabs.org/abs/2007ApJ...659..149M). doi: 10.1086/511807. [S2CID 15193438](https://doi.org/10.1086/511807).
25. E. Gaztañaga, K. Sravan Kumar, Swaraj Pradhan, and Michael Gabler, Gravitational bounce from the quantum exclusion principle, *Phys. Rev. D* **111**, 103537 (2025) This paper suggests a small but nonzero closed spatial curvature: $-0.07 \pm 0.02 \leq \Omega_k < 0$; DOI: <https://doi.org/10.1103/PhysRevD.111.103537>.
26. Moster, B. P. *et al.* Constraints on the relationship between stellar mass and halo mass at low and high redshift. *Astrophys. J.* **710**, 903–923 (2010).
27. Behroozi, P. S., Wechsler, R. H. & Conroy, C. The average star formation histories of galaxies in dark matter halos from $z = 0$ –8. *Astrophys. J.* **770**, 57 (2013).
28. P. van Dokkum et al., A galaxy lacking dark matter, *Nature* 555, 629 (2018).
29. Duncan Farrah, Kevin Croker, Michael Zevin et al., Observational Evidence for Cosmological Coupling of Black Holes and its Implications for an Astrophysical Source of Dark Energy, *The Astrophysical Journal Letters*, 944:L31, February 20, 2023.
30. A. Friedmann, Über die Möglichkeit einer Welt mit konstanter negativer Krümmung des Raumes, *Z. Physik* **21**, 326-332 (1924); On the Possibility of a universe with constant negative curvature of space, *General Relativity and Gravitation*, **31**, 1991 (2000).

³¹. J. C. Feng, Pisin Chen, Cosmological constant as an integration constant,
arXiv:2406.00932v3 [gr-qc] 13 Jun 2024.